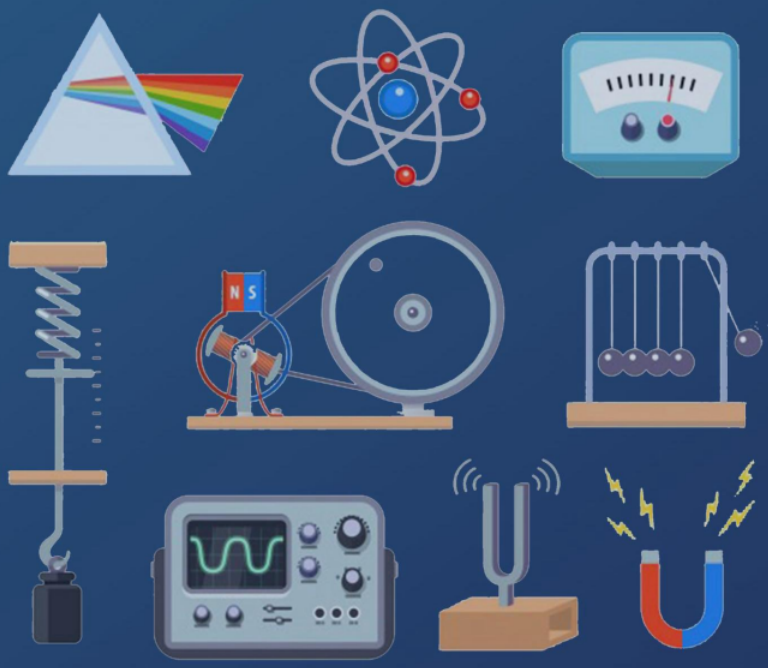
PHYSICS

Capacitor



SHEET

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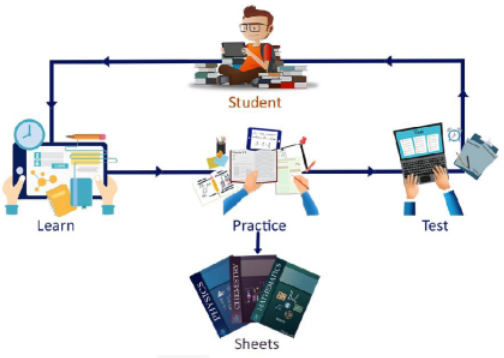
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Capacitor Introduction	1	15,16		11
2	Parallel Plate Capacitor	2	2		1,12
3	Series and Parallel combination	3-6			
4	Circuit Analysis- mesh and nodal analysis	7-9			2
5	Energy stored in capacitor	10,11			
6	Charge flow and heat loss	12-16	1,2	1-4	3
7	Equivalent capacitance and parallel plate	17	17,22	5,6	
8	Dielectric		3,18	7,8	
9	Differet combinations of dielectric	18-22		9	4
10	Electric force on dielectric slab			10,11	5
11	Transient state capacitor	23	4,9,10, 11-14		
12	Calculation of current at different state		5,6	12	6-9
13	Discharging of capacitor	24	7,8,19	13,14	
14	After full chapter	25	20,23	15	10

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Capacitor Introduction

Q1. The earth has Volume 'V' and Surface area 'A' then capacitance would be :

(A) $4\pi\epsilon_0 \frac{A}{V}$ (B) $4\pi\epsilon_0 \frac{V}{A}$

(C) $12\pi\epsilon_0 \frac{V}{A}$ (D) $12\pi\epsilon_0 \frac{A}{V}$

Parallel Plate Capacitor

Q2. **Statement–1 :** The capacitance of a capacitor depends on the shape, size and geometrical placing of the conductors and its medium between them.

Statement–2 : When a charge q passes through a battery of emf E from the negative terminal to an positive terminal, an amount qE of work is done by the battery.

(A) Statement-1 is True, Statement–2 is True; Statement-2 is a correct explanation for Statement-1

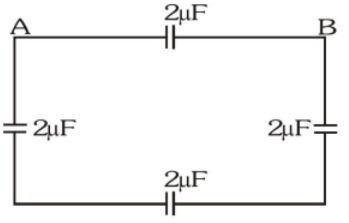
(B) Statement–1 is True, Statement–2 is True ; Statement–2 is NOT a correct explanation for Statement–1

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

Series and Parallel combination

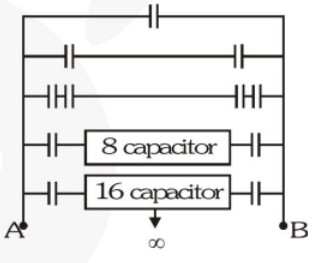
Q3. The equivalent capacitance between the points A and B in the given diagram is :



(A) 8 μF (B) 6 μF

(C) $\frac{8}{3}\mu F$ (D) $\frac{3}{8}\mu F$

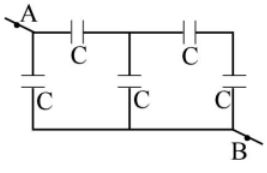
Q4. An infinite number of identical capacitors each of capacitance 1μ F are connected as in adjoining figure. Then the equivalent capacitance between A and B is :



(A) 1μ F (B) 2μ F

(C) $\frac{1}{2}\mu F$ (D) ∞

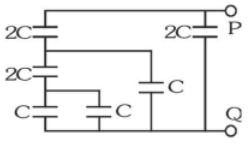
Q5. What is the equivalent capacitance of the system of capacitors between A & B as shown in the figure.



(A) $\frac{7}{6}C$ (B) 1.6 C

(C) C (D) None

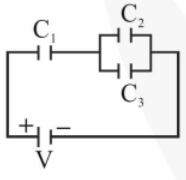
Q6. The value of equivalent capacitance of the combination shown in figure between the points P and Q is :–



(A) 3 C (B) 2 C (C) C (D) C/3

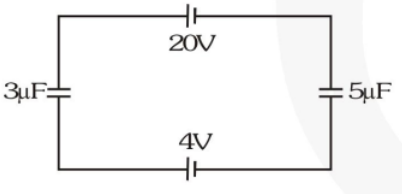
Circuit Analysis - mesh and nodal analysis

Q7. Capacitance $C_1 = 2C_2 = 2C_3$ and potential difference across C_1, C_2 & C_3 are V_1, V_2 & V_3 respectively then :-



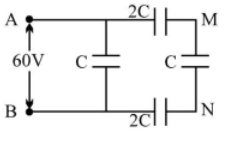
(A) $V_1 = V_2 = V_3$
(B) $V_1 = 2V_2 = 2V_3$
(C) $2V_1 = V_2 = V_3$
(D) $2V_1 = 2V_2 = V_3$

Q8. In the given circuit, the potential difference across 3 μF capacitor will be :



(A) 16 V (B) 10 V (C) 6 V (D) 4 V

Q9. In the circuit shown, a potential difference of 60V is applied across AB. The potential difference between the point M and N is

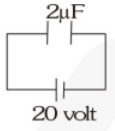


(A) 10 V (B) 15 V
(C) 20 V (D) 30 V

Energy stored in capacitor

Q10. A glass slab is put within the plates of a charged parallel plate condenser. Which of the following quantities does not change :
(A) energy of the condenser
(B) capacity
(C) intensity of electric field
(D) charge

Q11. In the figure a capacitor of capacitance 2μF is connected to a cell of emf 20 volt. The plates of the capacitor are drawn apart slowly to double the distance between them, The work done by the external agent on the plates is :

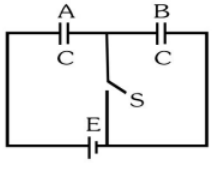


(A) – 200μJ (B) 200μJ
(C) 400μJ (D) – 400μJ

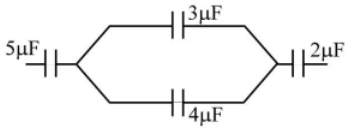
Charge Flow and Heat Loss

Q12. A parallel plate capacitor is charged with a battery and afterwards the battery is removed. If now with the help of insulating handles, the distance between plates is increased, then
(A) charge on capacitor increases and capacity decreases.
(B) potential difference between plates increases.
(C) capacity of capacitor increases.
(D) value of energy stored in capacitor decreases.

Q13. Consider the situation shown. The switch S is open for a long time and then closed. The charge flown through the battery when S is closed :
(A) CE
(B) CE / 2
(C) CE / 3
(D) CE / 4

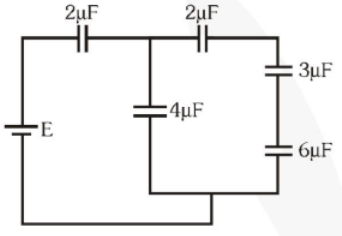


Q14. If charge on left plate of the 5μF capacitor in the circuit segment shown in the figure is –20μC, the charge on the right plate of 3μF capacitor is :-



(A) +8.57 μC (B) –8.57 μC
(C) +11.42 μC (D) –11.42 μC

Q15. In the given circuit if charge on 6μF capacitor is 10 μC, then charge on 4 μF capacitor will be :



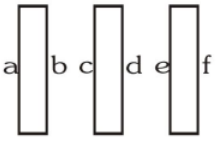
(A) 40 μC (B) 10 μC
(C) 20 μC (D) 30 μC

Q16. A capacitor of capacitance C is initially charged to a potential difference of V volt. Now it is connected to a battery of 2V Volt with opposite polarity. The ratio of heat generated to the final energy stored in the capacitor will be

(A) 1.75 (B) 2.25 (C) 2.5 (D) 1/2

Equivalent capacitance and parallel plate

Q17. Three parallel metallic plates, each of area A are kept as shown in the figure and charges Q_1, Q_2 and Q_3 are given to them. Edge effects are negligible. Calculate the charges on the two outermost surfaces 'a' and 'f'.



(A) $\frac{Q_1+Q_2+Q_3}{2}$ (B) $\frac{Q_1+Q_2+Q_3}{3}$

(C) $\frac{Q_1-Q_2+Q_3}{3}$ (D) $\frac{Q_1-Q_2+Q_3}{2}$

Differet Combinations of Dielectric

Q18. If the maximum circumference of a sphere is 2m, then capacitance of sphere in water would be nearly :

(Dielectric constant of water = 81)

(A) 270 pF (B) 2600 pF
(C) 28 pF (D) 2800 pF

Q19. **Statement–1:** Capacitor is filled with same thickness of dielectric (t < d) and conducting sheet one after another, then capacitance are C_1 and C_2 respectively then $C_1 < C_2$.

Statement–2: Capacitance is more in presence of metal sheet in compare to dielectric sheet as $K_{Conduct} > K_{dielectric}$.

(A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement-1

(B) Statement–1 is True, Statement–2 is True; Statement–2 is NOT a correct explanation for Statement–1

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

Q20. Distance between plates of a parallel plate capacitor is 'd' and area of each plate is A. When a slab of dielectric constant K and thickness t is placed between the plates, its capacity becomes:

(A) $\frac{\varepsilon_0 A}{\left[d+t\left\{1-\frac{1}{K}\right\}\right]}$ (B) $\frac{\varepsilon_0 A}{\left[d+t\left\{1+\frac{1}{K}\right\}\right]}$

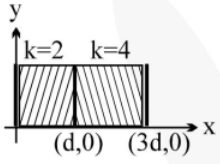
(C) $\frac{\varepsilon_0 A}{\left[d-t\left\{1+\frac{1}{K}\right\}\right]}$ (D) $\frac{\varepsilon_0 A}{\left[d-t\left\{1-\frac{1}{K}\right\}\right]}$

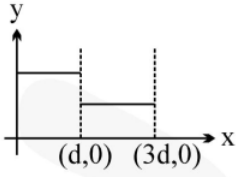
Q21. A capacitor stores 60μC charge when connected across a battery. When the gap between the plates is filled with a dielectric, a charge of 120μC flows through the battery. The dielectric constant of the material inserted is :

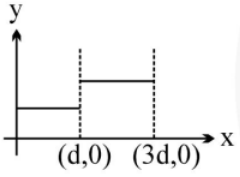
(A) 1 (B) 2

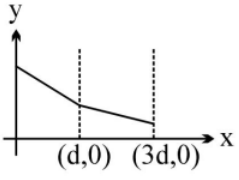
(C) 3 (D) none

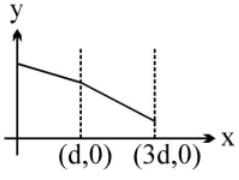
Q22. A parallel plate capacitor has two layers of dielectric as shown in figure. This capacitor is connected across a battery. The graph which shows the variation of electric field (E) and distance (x) from left plate is -



(A)


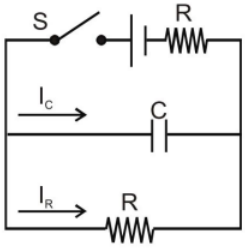
(B)


(C)


(D)


Transient state capacitor

Q23. Consider the circuit below, with an initially uncharged capacitor and two identical resistors. At the instant the switch is closed:



(A) $I_R = I_C = 0$

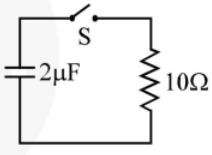
(B) $I_R = \varepsilon/2R$; $I_C = 0$

(C) $I_R = 0$; $I_C = \varepsilon/R$

(D) $I_R = \varepsilon/2R$; $I_C = \varepsilon/R$

Discharging of Capacitor

Q24. In the R–C circuit shown in the figure the total energy of 3.6×10^{-3} J is dissipated in the 10 Ω resistor when the switch S is closed. The initial charge on the capacitor is



(A) 60 μC (B) 120 μC

(C) $60\sqrt{2}$ μC (D) $\frac{60}{\sqrt{2}}$ μC

After Full Chapter

Q25. Three capacitors 2 μF, 3 μF and 5 μF can withstand voltages to 3V, 2V and 1V respectively. Their series combination can withstand a maximum voltage equal to

(A) 5 Volts

(B) (31/6) Volts

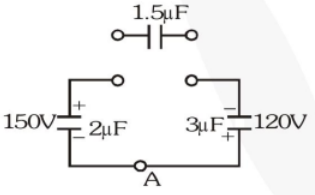
(C) (26/5) Volts

(D) None

ONE OR MORE THAN ONE CORRECT TYPE

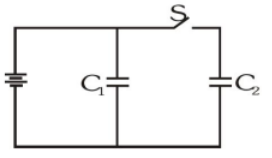
*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Two capacitors of 2μF and 3μF are charged to 150 volt and 120 volt respectively. The plates of capacitor are connected as shown in the figure. A discharged capacitor of capacity 1.5μF falls to the free ends of the wire. Then –



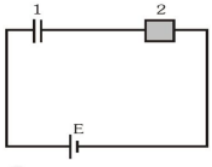
(A) charge on the 1.5μF capacitor is 180μC
(B) charge on the 2μF capacitor is 120 μC
(C) charge flows through A from right to left
(D) charge flows through A from left to right.

Q2. Two capacitors of equal capacitance (C_1=C_2) are shown in the figure. Initially, while the switch S is open, one of the capacitors is uncharged and the other carries charge Q_0. The energy stored in the charged capacitor is U_0. Sometimes after the switch is closed, the capacitors C_1 and C_2 carry charges Q_1 and Q_2 respectively; the voltage across the capacitors are V_1 and V_2; and the energies stored in the capacitors are U_1 and U_2. Which of the following statements is correct ?



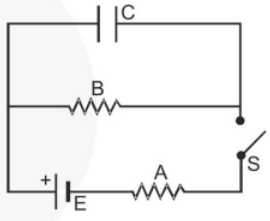
(A) $Q_0 = \frac{1}{2}(Q_1 + Q_2)$ (B) $Q_1 = Q_2$
(C) $V_1 = V_2$ (D) $U_1 = U_2$

Q3. Two identical capacitors 1 and 2 are connected in series to a battery as shown in figure. Capacitor 2 contains a dielectric slab of dielectric constant k as shown. Q_1 and Q_2 are the charges stored in the capacitors. Now the dielectric slab is removed and the corresponding charges are Q'_1 and Q'_2. Then-



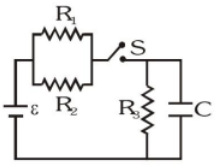
(A) $\frac{Q'_1}{Q_1} = \frac{k+1}{k}$ (B) $\frac{Q'_2}{Q_2} = \frac{k+1}{2}$
(C) $\frac{Q'_2}{Q_2} = \frac{k+1}{2k}$ (D) $\frac{Q'_1}{Q_1} = \frac{k}{2}$

Q4. In the circuit shown, A and B are equal resistances. When S is closed, the capacitor C charges from the cell of emf E and reaches a steady state



(A) During charging, more heat is produced in A than in B.
(B) In the steady state, heat is produced at the same rate in A and B.
(C) In the steady state, energy stored in C is 1/4 CE^2.
(D) In the steady state, energy stored in C is 1/8 CE^2

Q5. The circuit shown in the figure consists of a battery of emf ε=10 V; a capacitor of capacitance C =1.0 μF and three resistor of values R_1 = 2Ω, R_2 = 2Ω and R_3 = 1Ω. Initially the capacitor is completely uncharged and the switch S is open. The switch S is closed at t = 0.



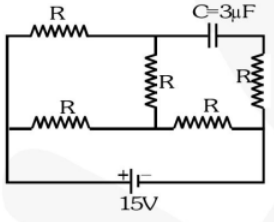
(A) The current through resistor R_3 at the moment the switch closed is zero

(B) The current through resistor R_3 a long time after the switch closed is 5A.

(C) The ratio of current through R_1 and R_2 is always constant

(D) The maximum charge on the capacitor during the operation is 5μC

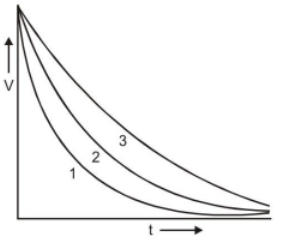
Q6. In the circuit shown, the cell is ideal, with emf = 15V. Each resistance is of 3Ω. After long time The potential difference across the capacitor is



(A) zero (B) 9V

(C) 12V (D) 15V

Q7. Three identical capacitors A, B and C are charged to the same potential and then made to discharge through three resistances R_A, R_B and R_C, where $R_A > R_B > R_C$. Their potential differences (V) are plotted against time t, giving the curves 1, 2 and 3. The relations between A, B, C and 1, 2, 3 is/are -



(A) 1 → A (B) 2 → B

(C) 1 → C (D) 3 → A

Q8. Capacitors $C_1 = 1\mu F$ and $C_2 = 2\mu F$ are separately charged from the same battery. They are then allowed to discharge separately through equal resistors–

(A) The currents in the two discharging circuits at t = 0 is zero.

(B) The currents in the two discharging circuits at t = 0 are equal but not zero.

(C) The currents in the two discharging circuits at t = 0 are unequal.

(D) C_1 loses 50% of its initial charges sooner than C_2 loses 50% of its initial charge.

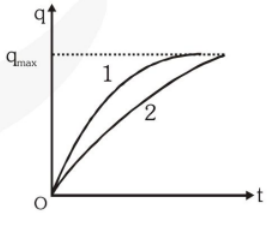
PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 9 & 10

The charge across the capacitor in two different RC circuits 1 and 2 are plotted as shown in figure.

Q9. Choose the correct statement (s) related to the two circuits.



(A) Both the capacitors are charged to the same charge.

(B) The emf's of cells in both the circuit are equal.

(C) The emf's of the cells may be different.

(D) The emf E_1 is more than E_2

Q10. Identify the correct statement(s) related to the R_1, R_2, C_1 and C_2 of the two RC circuits if time constant of second circuit is more than first circuit.

(A) $R_1 > R_2$ if $E_1 = E_2$

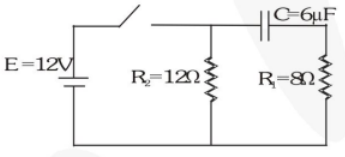
(B) $C_1 < C_2$ if $E_1 = E_2$

(C) $R_1C_1 > R_2C_2$

(D) $\frac{R_1}{R_2} < \frac{C_2}{C_1}$

Paragraph for Question No. 11 & 14

The circuit contains ideal battery E and other elements arranged as shown. The capacitor is initially uncharged and switch S is closed at t = 0, (use $e^2 = 7.4$):



Q11. Time constant of the circuit is:

(A) 48 μs (B) 28.8 μs

(C) 72 μs (D) 120 μs

Q12. The potential difference across the capacitor in volts, after two time constants, is :

(A) 2 (B) 7.6

(C) 10.4 (D) 12

Q13. The potential difference across resistor R_1 in volts after two time constants, is :

(A) 1.6 (B) 7.6

(C) 10 (D) 12

Q14. The potential difference across resistor R_2 in volts after two time constants, is:

(A) 2 (B) 7.6

(C) 10 (D) 12

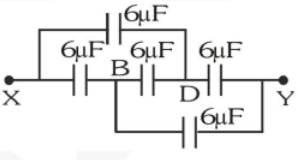
NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q15. A conductor gets a charge of 50 μC when it is connected to a battery of e.m.f. 5 V. Calculate capacity of the conductor.

Q16. Two conductors having capacities 2μF and 5μF and potentials 2 volt and 10 volt respectively. Calculate the ratio of their charges after connecting by a wire.

Q17. What is the effective capacitance between points X and Y ?



Q18. The capacity of a spherical capacitor in air is 50 μF and on immersing it into oil it becames 110 μF. Calculate the dielectric constant of oil.

Q19. The plates of a capacitor are charged to a potential difference of 100 V and then connected across a resister. The potential difference across the capacitor decays exponentially with respect to time. After one second the potential difference between the plates of the capacitor is 80 V. What is the fraction of the stored energy which has been dissipated ?

Q20. A parallel-plate capacitor is formed by two plates, each of area 100 cm^2, separated by a distance of 1mm. A dielectric of dielectric constant 5.0 and dielectric strength 1.9×10^7 V/m is filled between the plates. Find the maximum charge that can be stored on the capacitor without causing any dielectric breakdown.

SUBJECTIVE TYPE

Q21. If the distance between the plates of a parallel plate capacitor is made half and the area of plates is doubled then what will be the capacitance.

Q22. An infinite number of capacitors of capacitance C, 4C, 16C ... ∞ are connected in series then what will be their resultant capacitance ?

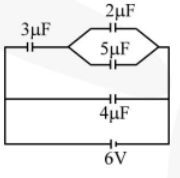
Q23. A parallel plate capacitor of capacity C is connected to a battery and is charged to potential V, another condenser of capacity 2C is connected to another battery and is charged to potential 2V the charging batteries are removed and now the condensers are connected in parallel in such a way that the positive plate of one is connected to negative plate of another. Calculate final energy of the system.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

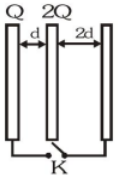
Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. In the circuit shown in figure, the ratio of charges on 5μF and 4μF capacitor is :-



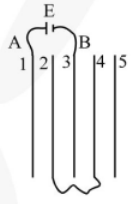
(A) 4/5 (B) 3/5
(C) 3/8 (D) 1/2

Q2. Three large plates are arranged as shown. How much charge will flow through the key k if it is closed ?



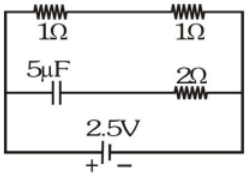
(A) $\frac{5Q}{6}$ (B) $\frac{4Q}{3}$
(C) $\frac{3Q}{2}$ (D) None of these

Q3. Five conducting parallel plates having area A and separation between them d, are placed as shown in the figure. Plate number 2 and 4 are connected wire and between point A and B, a cell of emf E is connected. The charge flown through the cell is



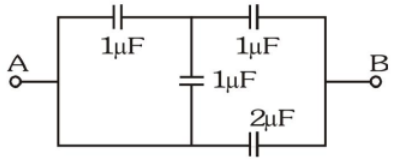
(A) $\frac{3}{4}\frac{\varepsilon_0 AE}{d}$ (B) $\frac{2}{3}\frac{\varepsilon_0 AE}{d}$
(C) $\frac{4\varepsilon_0 AE}{d}$ (D) $\frac{\varepsilon_0 AE}{2d}$

Q4. A capacitor of capacitance 5 μF is connected as shown in the figure The internal resistance of the cell is 0.5 Ω. The amount of charge on the capacitor plate is :-



(A) 0 μC (B) 5 μC
(C) 10 μC (D) 25 μC

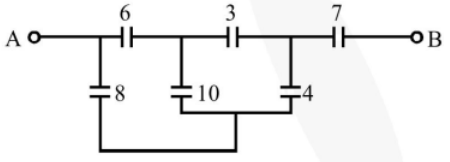
Q5. The equivalent capacitance of the circuit shown, between points A and B will be



(A) $\frac{2}{3}\mu F$ (B) $\frac{5}{3}\mu F$

(C) $\frac{8}{3}\mu F$ (D) $\frac{7}{3}\mu F$

Q6. In the circuit diagram shown all the capacitors are in mF. The equivalent capacitance between points A & B is (in mF)



(A) 14/5 (B) 7.5

(C) 3/7 (D) None of these

Q7. Two identical parallel plate capacitors are connected in series and then joined with a battery of 100V. A sheet of dielectric constant 4.0 is inserted between the plates of second capacitor. The potential difference across the capacitors will be respectively

(A) 50 V, 50V (B) 80 V, 20 V

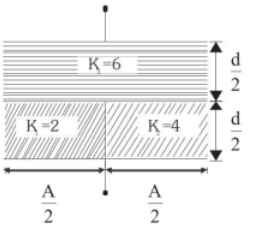
(C) 20V, 80V (D) 75 V, 25 V

Q8. Two parallel plate capacitors whose capacities are C and 2 C respectively, are joined in parallel. These are charged by V potential difference. If the battery is now removed and a dielectric of dielectric constant K is filled in between the plates of the capacitor C, then what will be the potential difference across each capacitor ?

(A) $\frac{V}{K+2}$ (B) $\frac{2V}{K+2}$

(C) $\frac{3V}{K+2}$ (D) $\frac{2+K}{3V}$

Q9. A parallel plate capacitor of capacitance C (without dielectrics) is filled by dielectric slabs as shown in figure. Then the new capacitance of the capacitor is



(A) 3.9 C

(B) 4 C

(C) 2.4 C

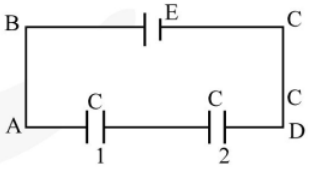
(D) 3 C

Q10. A parallel plate capacitor of plate area A and plate separation d is charged to potential V and then the battery is disconnected. A slab of dielectric constant k is then inserted between the plates of the capacitors so as to fill the space between the plates. If Q, E and W denote respectively, the magnitude of charge on each plate, the electric field between the plates (after the slab is inserted) and work done on the system in question in the process of inserting the slab, then state incorrect relation from the following :-

(A) $Q=\frac{\varepsilon_0 AV}{d}$ (B) $W=\frac{\varepsilon_0 AV^2}{2kd}$

(C) $E=\frac{V}{kd}$ (D) $W=\frac{\varepsilon_0 AV^2}{2d}\left(1-\frac{1}{k}\right)$

Q11. In the adjoining figure, capacitor (1) and (2) have a capacitance 'C' each. When the dielectric of dielectric consatnt K is inserted between the plates of one of the capacitor, the total charge flowing through battery is-



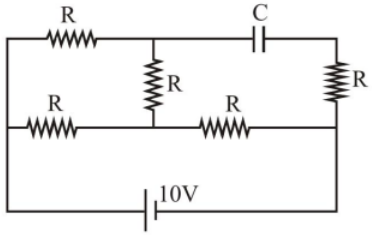
(A) $\frac{KCE}{K+1}$ from B to C

(B) $\frac{KCE}{K+1}$ from C to B

(C) $\frac{(K-1)CE}{2(K+1)}$ from B to C

(D) $\frac{(K-1)CE}{2(K+1)}$ from C to B

Q12. Find the potential difference across the capacitor in volts.



(A) 4V (B) 2V
(C) 1 V (D) 8V

Q13. A capacitor of capacitance C is charged by a battery whose internal resistance is R. The time after which potential difference across resistor becomes n times to that across capacitor is -

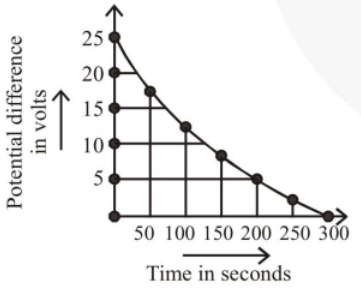
(A) $RC\ell n\left(\frac{1+n}{n}\right)$

(B) $RC\ell n\left(\frac{n}{1+n}\right)$

(C) $RC\ell n\left(\frac{n}{n-1}\right)$

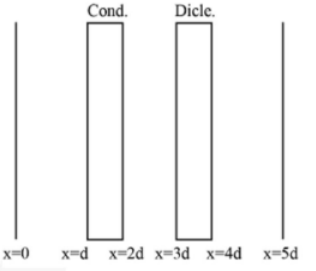
(D) $RC\ell n(1+n)$

Q14. The figure shows an experimental plot discharging of a capacitor in an RC circuit. The time constant t of this circuit lies between -

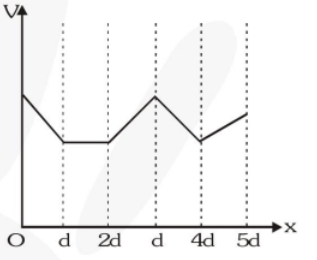


(A) 150 sec and 200 sec
(B) 0 and 50 sec
(C) 50 sec and 100 sec
(D) 100 sec and 150 sec

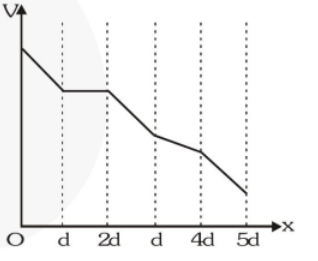
Q15. The distance between plates of a parallel plate capacitor is 5d. The positively charged plate is at x=0 and negatively charged plates is at x=5d. Two slabs one of conductor and the other of a dielectric of same thickness d are inserted between the plates as shown in figure. Potential (V) versus distance x graph will be



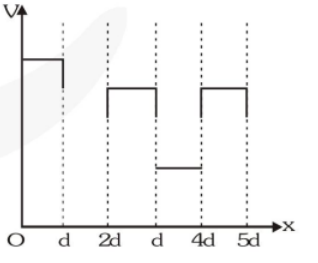
(A)



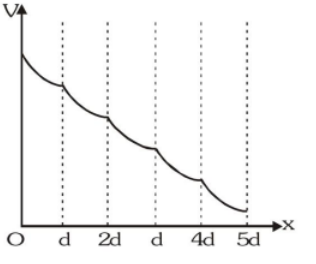
(B)



(C)



(D)



ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Three long concentric conducting cylindrical shells have radii R, 2R and $2\sqrt{2}$ R. Inner and outer shells are connected to each other. The capacitance across middle and inner shells per unit length is:

(A) $\frac{\frac{1}{3}\epsilon_0}{l\text{n}\,2}$ (B) $\frac{6\pi\epsilon_0}{l\text{n}\,2}$

(C) $\frac{\pi\epsilon_0}{2l\text{n}\,2}$ (D) None

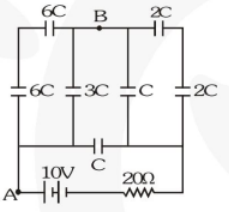
Q2. For the circuit shown here, the potential difference between points A and B is

(A) 2.5 V

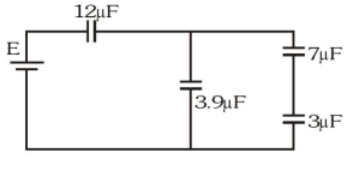
(B) 7.5 V

(C) 10 V

(D) Zero



Q3. Four capacitors and a battery are connected as shown. The potential drop across the 7μF capacitor is 6V. Then the



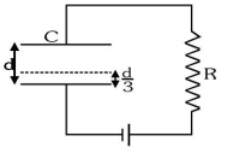
(A) potential difference across the 3μF capacitor is 10V

(B) charge on the 3μF capacitor is 42μC

(C) e.m.f. of the battery is 30V

(D) potential difference across the 12μF capacitor is 10V.

Q4. A parallel plate capacitor C with plates of unit area and separation d is filled with a liquid of dielectric constant K = 2. The level of liquid is $\frac{d}{3}$ initially. Suppose the liquid level decreases at a constant speed v the time constant as a function of time t is :



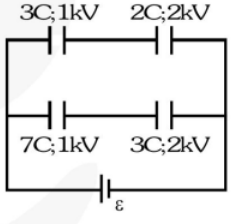
(A) $\frac{6\varepsilon_0 R}{5d+3vt}$

(B) $\frac{(15d+9vt)\varepsilon_0 R}{2d^2-3dvt-9v^2t^2}$

(C) $\frac{6\varepsilon_0 R}{5d-3vt}$

(D) $\frac{(15d-9vt)\varepsilon_0 R}{2d^2+3dvt-9v^2t^2}$

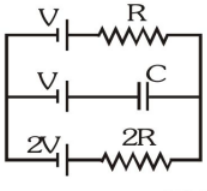
Q5. The diagram shows four capacitors with capacitance and break down voltages as mentioned. What should be the maximum value of the external emf source such that no capacitor breaks down?



(A) 2.5 kV (B) 10/3kV

(C) 3 kV (D) 1 kV

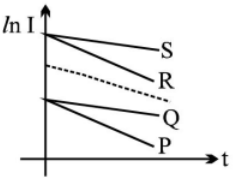
Q6. In the given circuit, with steady current, the potential drop across the capacitor must be:-



(A) V (B) V/2

(C) V/3 (D) 2V/3

Q7. In an RC circuit while charging, the graph of $\ell n\, I$ versus time is as shown by the dotted line in the adjoining diagram where I is the current. When the value of the resistance is doubled, which of the solid curves best represents the variation of $\ell n\, I$ versus time?

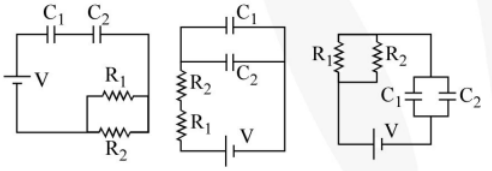


(A) P (B) Q (C) R (D) S

Q8. Given :

$R_1 = 1\Omega$, $R_2 = 2\Omega$, $C_1 = 2\mu F$, $C_2 = 4\mu F$.

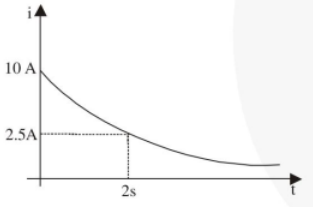
The time constants (in μS) for the circuits I, II, III are respectively



(A) 18, 8/9, 4 (B) 18, 4, 8/9

(C) 4, 8/9, 18 (D) 8/9, 18, 4

Q9. The figure shows, graph of the current of a discharging circuit of a capacitor through a resistor of resistance 10Ω:



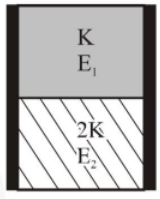
(A) The initial potential difference across the capacitor is 100 volt.

(B) The capacitance of the capacitor is $\frac{1}{10\ell n2}$ F.

(C) The total heat produced in the circuit will be $\frac{500}{\ell n2}$ joules

(D) The thermal power in the resistor will decreases with a time constant $\frac{1}{2\ell n2}$ second.

Q10. A parallel plate capacitor is connected from a cell and then isolated from it. Two dielectric slabs of dielectric constant K and 2K are now introduce in the region between upper half and lower half of the plate (as shown in figure). The electric field intensity in upper half of dielectric is E_1 and lower half is E_2 then -



(A) $E_1 = 2E_2$

(B) Electrostatic potential energy of upper half is less than that of lower half

(C) Induced charges on both slabs are same

(D) Charge distribution on the plates re-mains same after insertion of dielectric.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q11. Two uniformly charged spherical drops at potential V coalapse to form a larger drop. If capacity of each smaller drop is C then find capacity and potential of larger drop.

Q12. A cylindrical capacitor has two co-axial cylinders of length 15 cm and radii 1.5 cm and 1.4 cm. The outer cylinder is earthed and the inner cylinder is given a charge of 3.5 μC. Determine the capacitance of the system and the potential of the inner cylinder.

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	B	C	B	B	A	A	B	D	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	B	B	A	A	B	A	D	A	D
21	**22**	**23**	**24**	**25**					
C	A	C	B	B					

Ex-1A

1. A,B,C **2.** A,B,C,D **3.** C **4.** A,B,D **5.** A,B,C,D

6. C **7.** B,C,D **8.** B,D **9.** A, C **10.** D

11. A **12.** C **13.** A **14.** D **15.** 10 μF

16. 2/5 **17.** 6μF. **18.** 2.2 **19.** $\frac{9}{25}$ **20.** 8.4×10^{-6} C.

21. $C_2 = 4C_1$ **22.** $C_{eq} = \frac{3}{4}C$ **23.** $= \frac{3}{2}CV^2$

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	A	B	C	C	A	B	C	A	B
11	**12**	**13**	**14**	**15**					
D	D	A	D	B					

Ex-2A

1. B **2.** A **3.** B,C,D **4.** A **5.** A

6. C **7.** B **8.** D **9.** A,B,C,D **10.** B

11. $2^{2/3}$ V **12.** 2.89×10^4 V

-: Important Notes & Formulas :-